5.9 Cutter holding devices

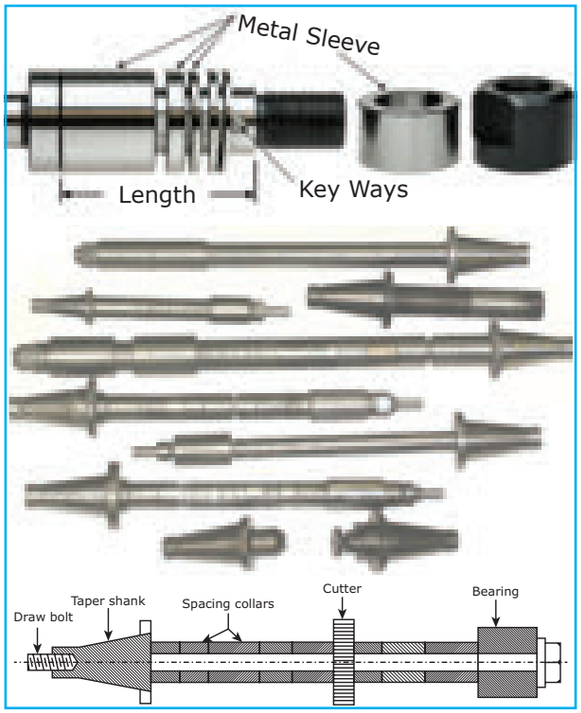
Depending on the design of the cutter there are several methods of supporting milling cutters on the machine spindle.

1. Arbor 2. Collet 3. Adapter
4. Screwed on cutters

Arbor

Milling cutters with central holes are mounted and keyed on a shaft called arbor. There are three different types of arbor namely pilot end arbor, 'A' type arbor and stub arbor.

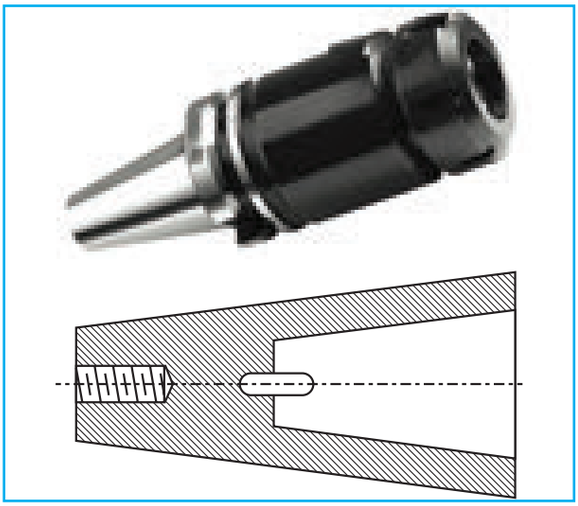
The arbours are made with taper shanks for correct alignment with the machine spindle the left side of the arbor is threaded internally to receive a draw bolt. This draw bolt connects the arbor with the spindle. A long key way is cut on the entire length of the arbor. Cutters are mounted at desired positions on the arbor by placing spacing collars between them. The spindle rotation is transmitted to the arbor and the cutter is rotated. An arbor is illustrated in figure



Arbor

Collet

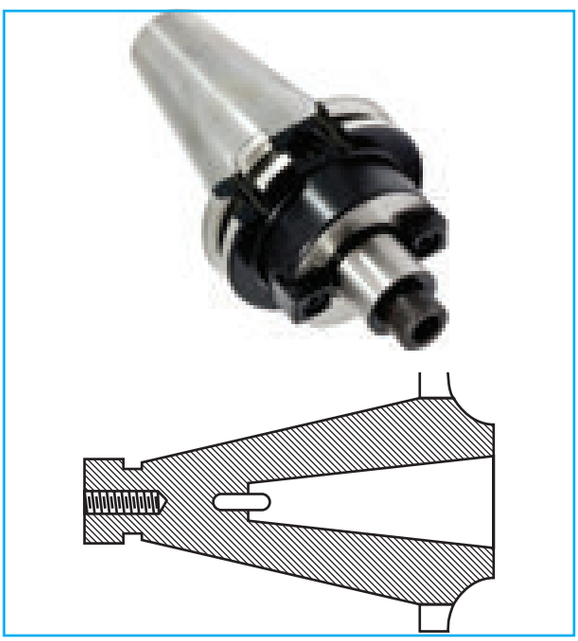
It is form of sleeve bushing used to hold arbor or cutters having a smaller shank than the spindle taper. Collets are connected to the spindle by a draw bolt and the rotary motion is transmitted to the cutters.



Collet

Adapters

Milling cutters having shanks are generally mounted on adapter. The outside taper of the adapter confirms to the taper hole of the spindle. The shank of the cutter fits into the taper hole of the adapter.



Adapter

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